

ML Expense Analyzer Project Report

Project: ML Expense Analyzer

Version: 1.0.0

Description: Machine Learning project for personal expense analysis and prediction

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Dataset Information:

- Total Records: 61
- Date Range: 2026-01-01 to 2026-01-31
- Total Expenses: \$2,238.50
- Average Transaction: \$36.70
- Unique Categories: 8

Machine Learning Components:

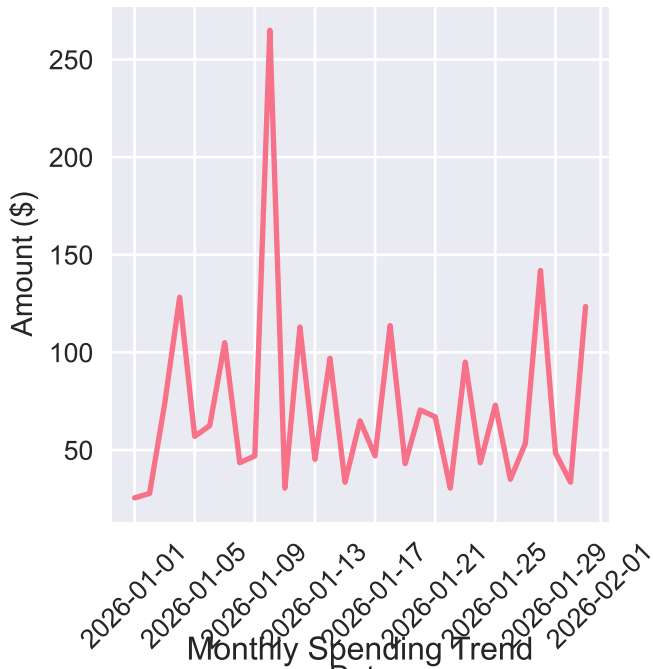
- Supervised Learning: Expense Categorization (Classification)
- Supervised Learning: Future Expense Prediction (Regression)
- Unsupervised Learning: Spending Pattern Discovery (Clustering)
- Anomaly Detection: Fraudulent Transaction Identification

Key Features:

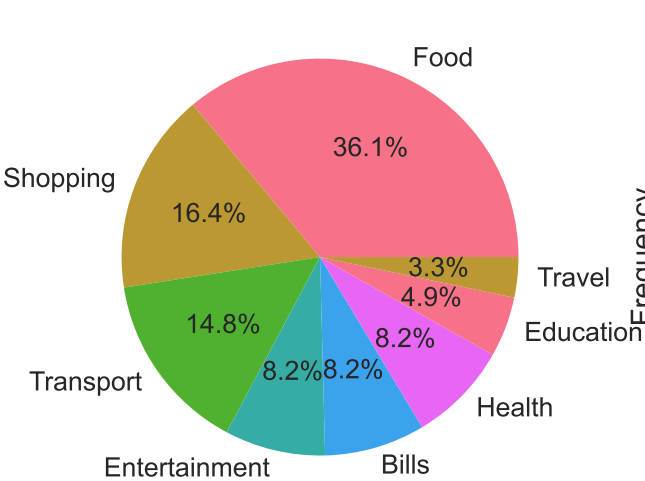
- Automated expense categorization
- Spending trend analysis and forecasting
- Financial anomaly detection
- Interactive visualizations
- Exportable reports and predictions

Data Analysis and Feature Engineering

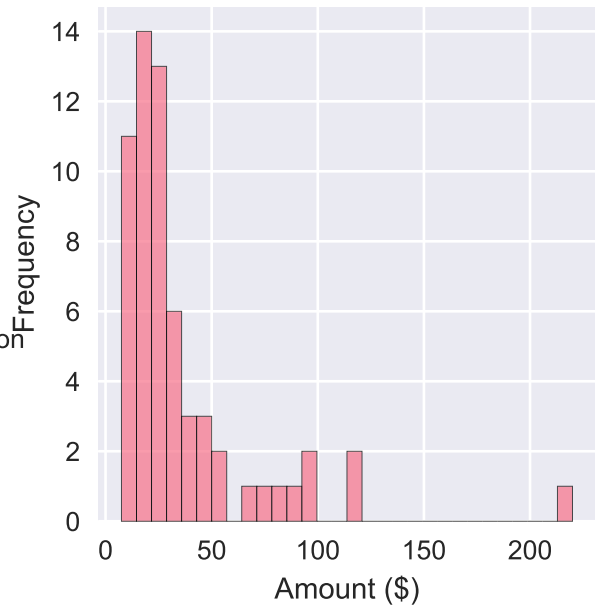
Daily Expenses Over Time



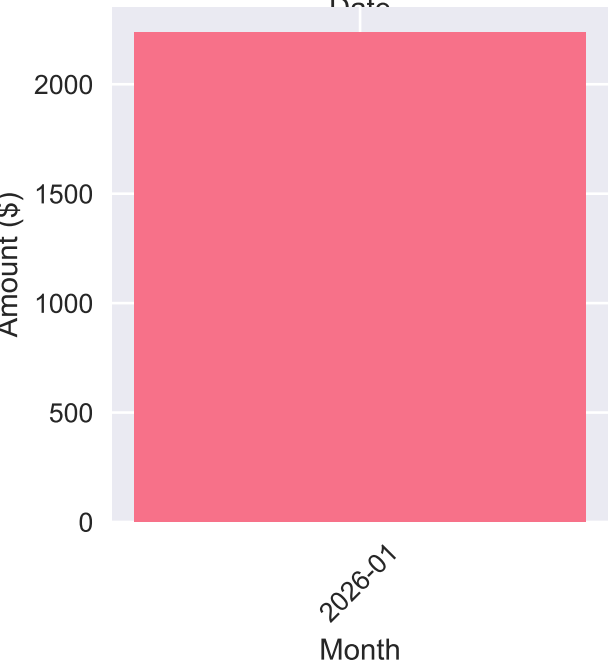
Expense Distribution by Category



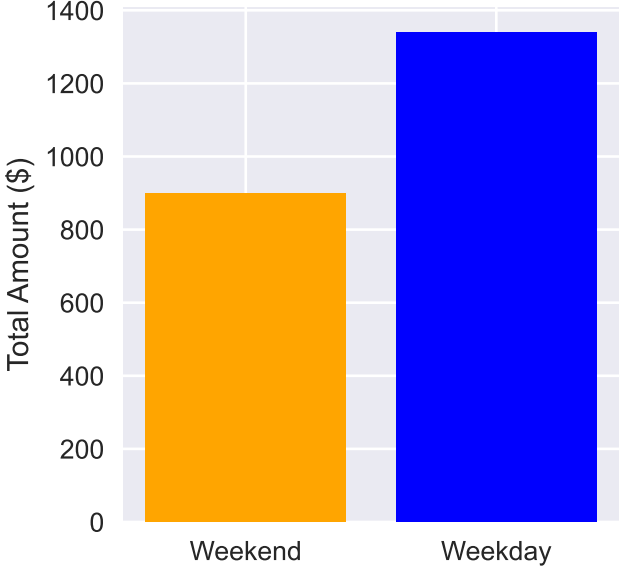
Transaction Amount Distribution



Monthly Spending Trend



Weekend vs Weekday Spending



Feature Engineering Summary:

- Original Features: 8
- Engineered Features: 74
- Total Features: 82

Feature Types:

- Time-based: Hour, day, month, quarter, weekend flags
- Amount-based: Log transform, z-score, rolling statistics
- Text-based: Description length, word count, sentiment
- Categorical: Encoded category, currency, payment method
- Interaction: Amount-time, category-amount combinations

Model Performance Summary

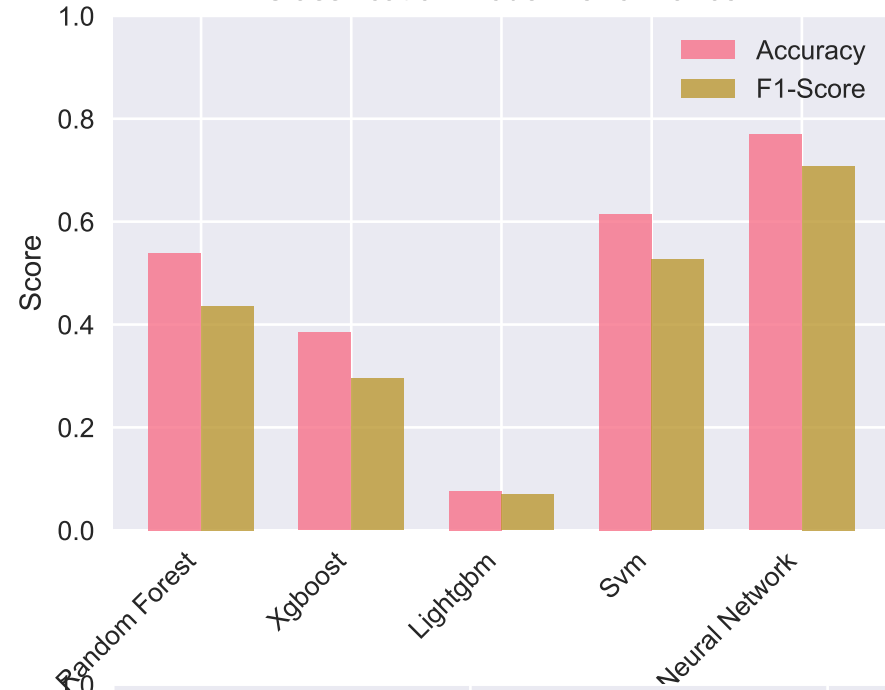
Classification Models		
Model	Accuracy	F1-Score
Random Forest	0.538	0.436
Xgboost	0.385	0.295
Lightgbm	0.077	0.070
Svm	0.615	0.527
Neural Network	0.769	0.709
Regression Models		
Model	R ² Score	RMSE
Linear Regression	-0.086	39.972
Ridge	-0.086	39.972
Lasso	-0.625	48.881
Gradient Boosting	-0.060	39.478
Xgboost Regressor	0.254	33.124
Clustering Models		
Model	Clusters	Silhouette Score
Kmeans	5	0.058
Dbscan	0	N/A
Anomaly Detection Models		
Model	Anomaly Rate	Count
Isolation Forest	9.84%	6
One Class Svm	31.15%	19

- Model Performance Metrics:
- Classification: Accuracy (correct predictions), F1-Score (balance of precision & recall)
 - Regression: R² (variance explained), RMSE (prediction error)
 - Clustering: Number of clusters found, Silhouette Score (cluster quality)
 - Anomaly Detection: Percentage of transactions flagged as unusual

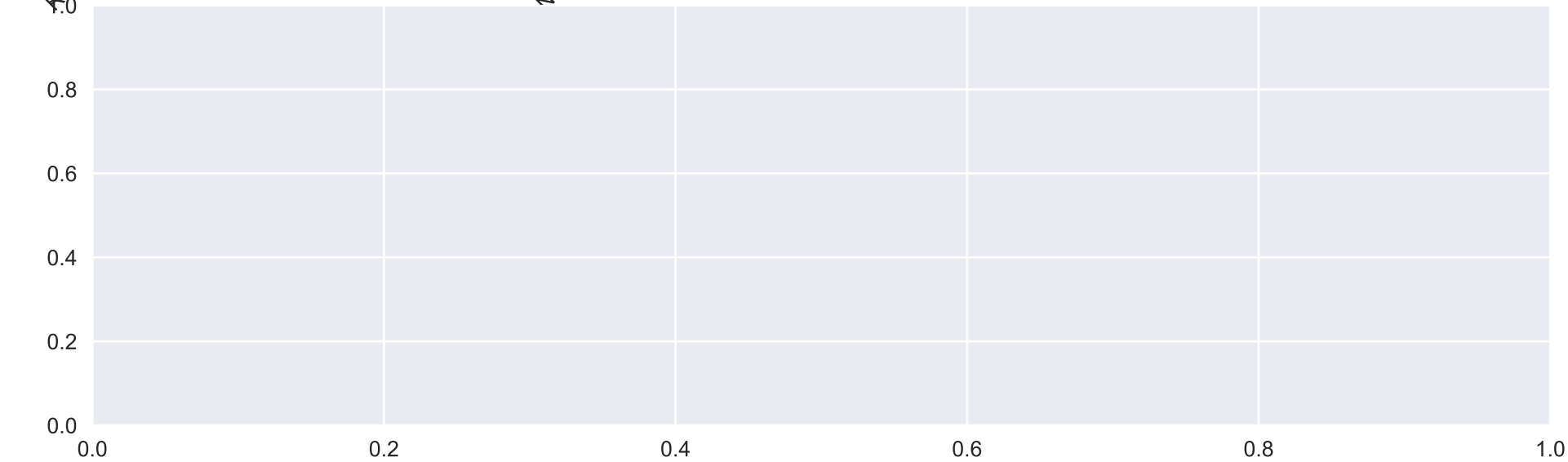
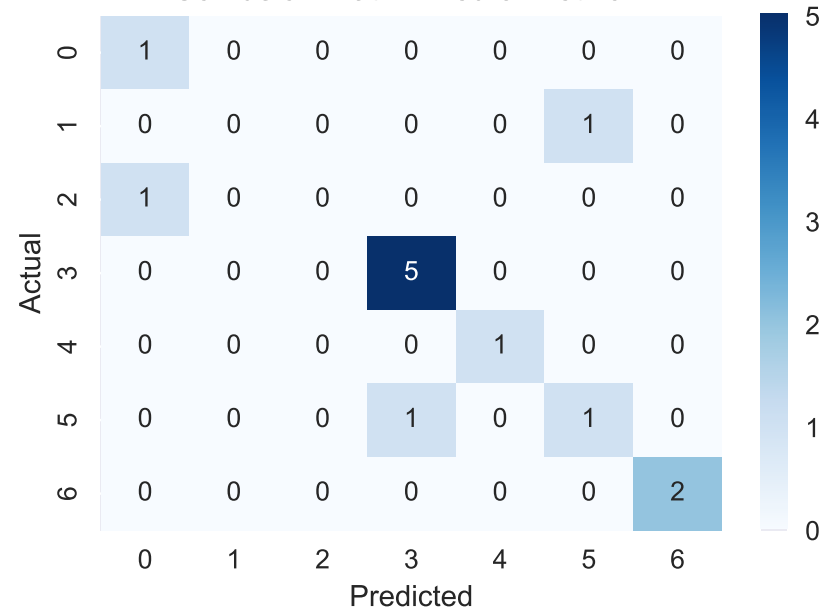
Note: Higher values indicate better performance for all metrics except RMSE (lower is better).

Classification Model Details

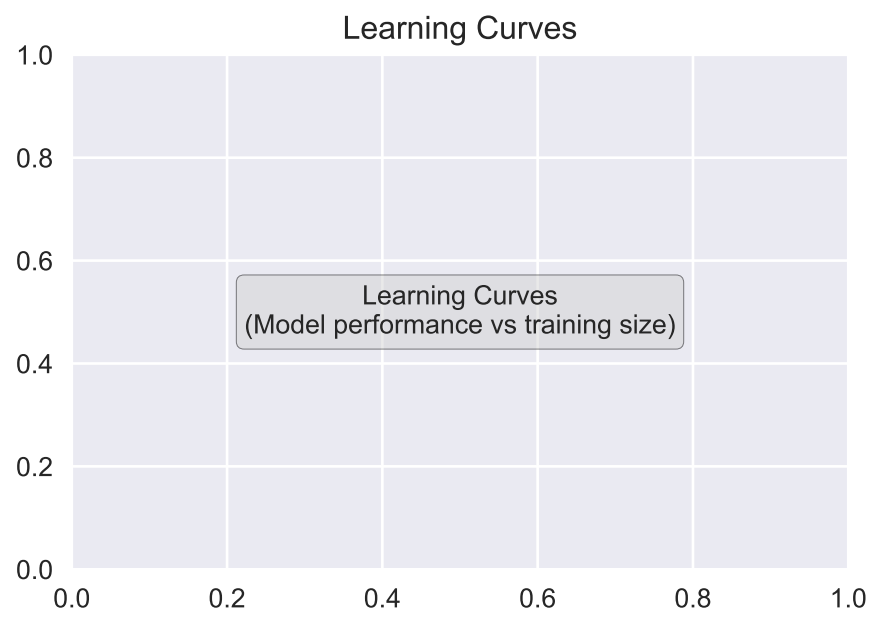
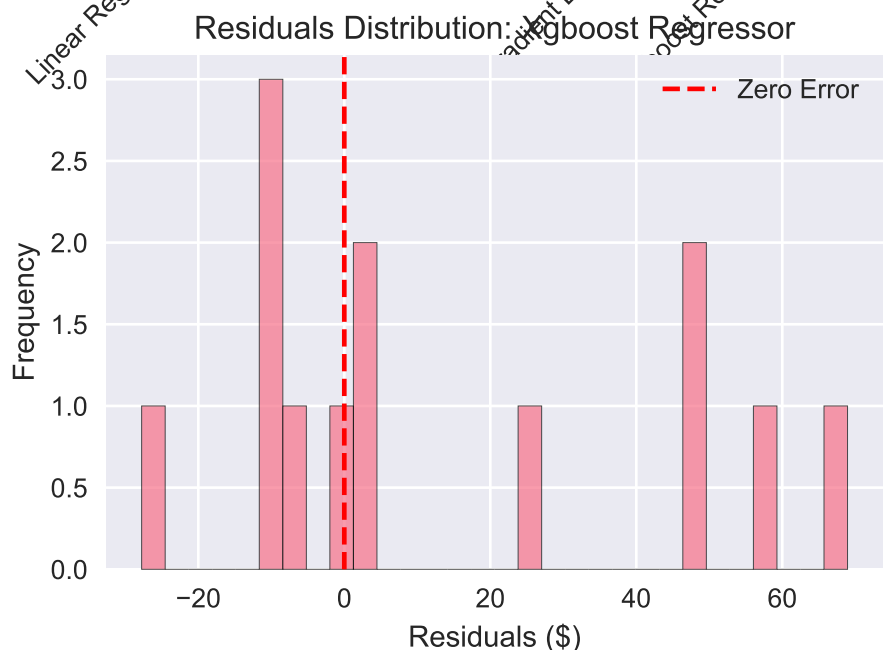
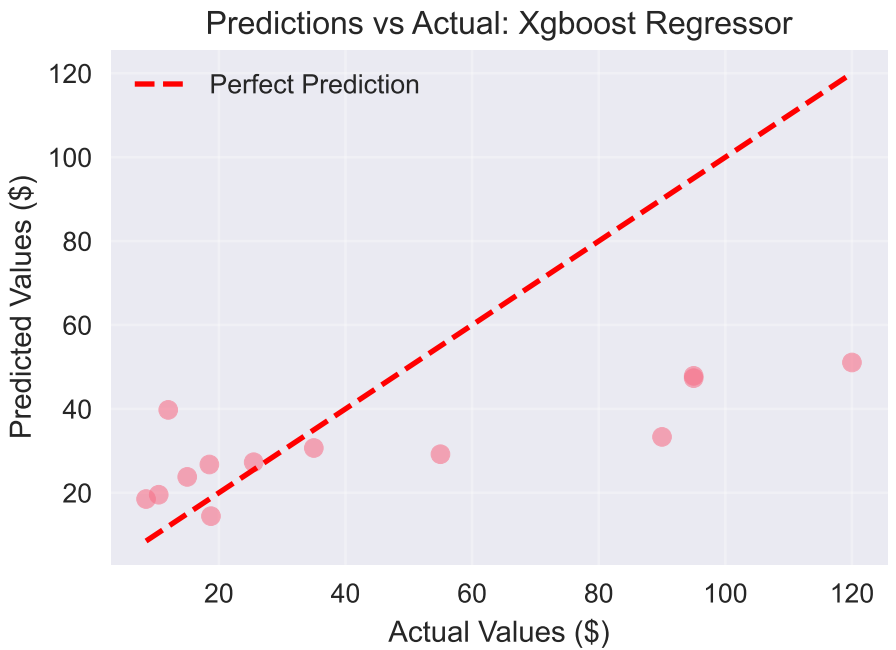
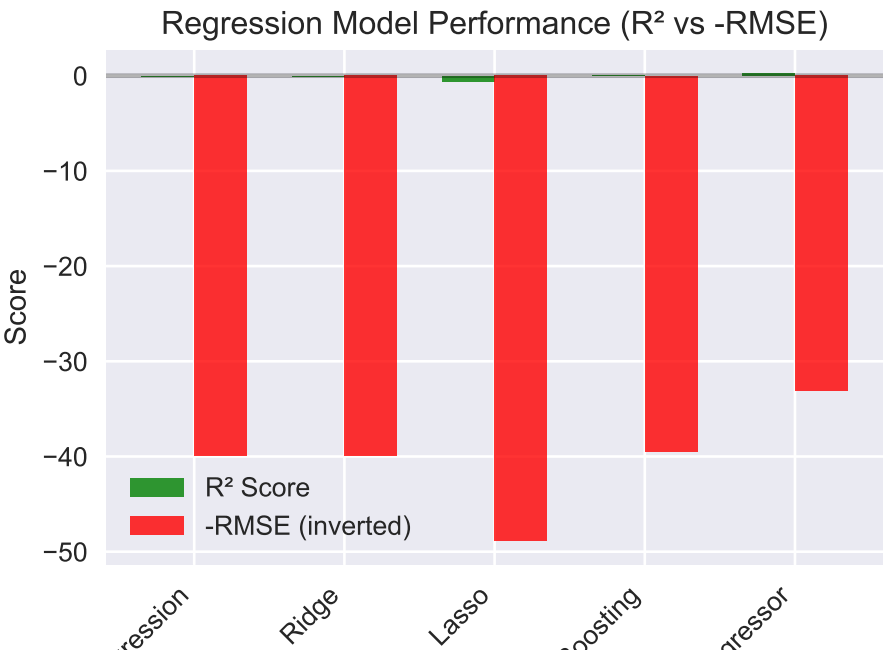
Classification Model Performance



Confusion Matrix: Neural Network

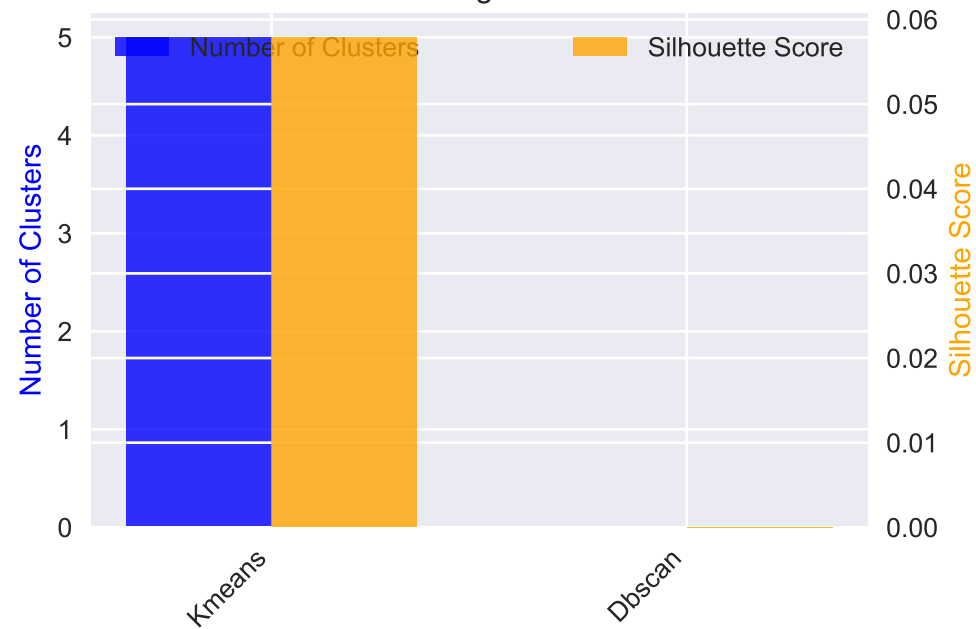


Regression Model Details

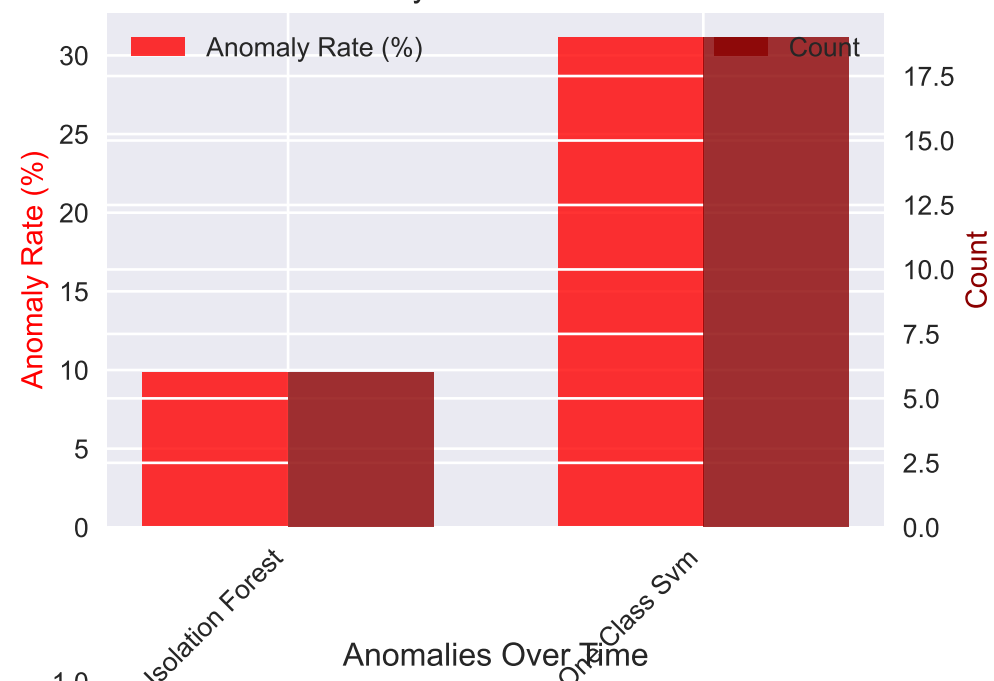


Unsupervised Learning Results

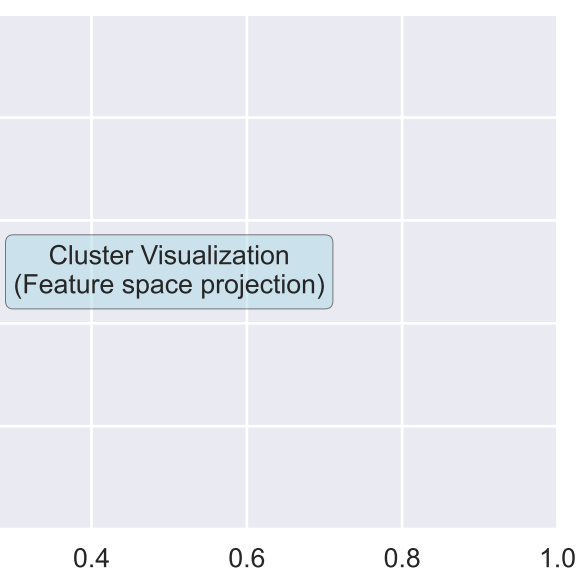
Clustering Results



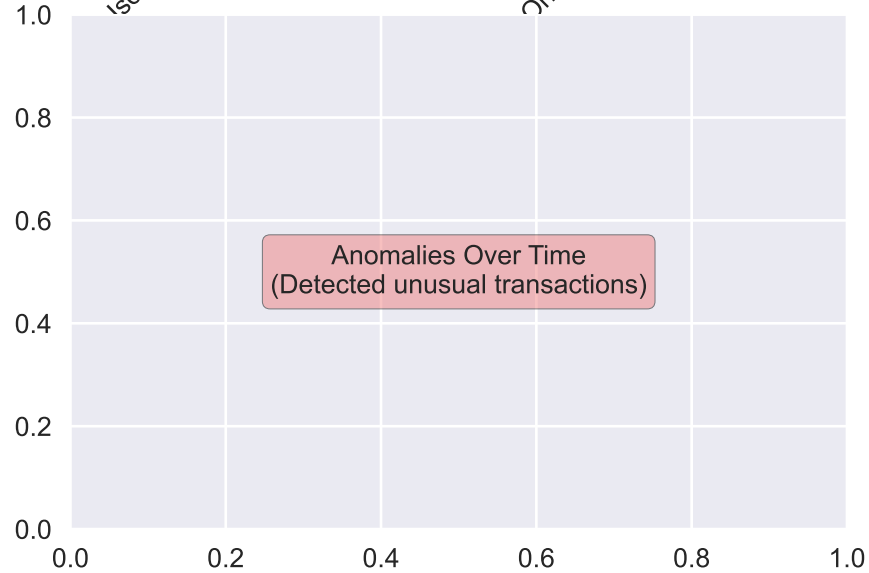
Anomaly Detection Results



Cluster Visualization

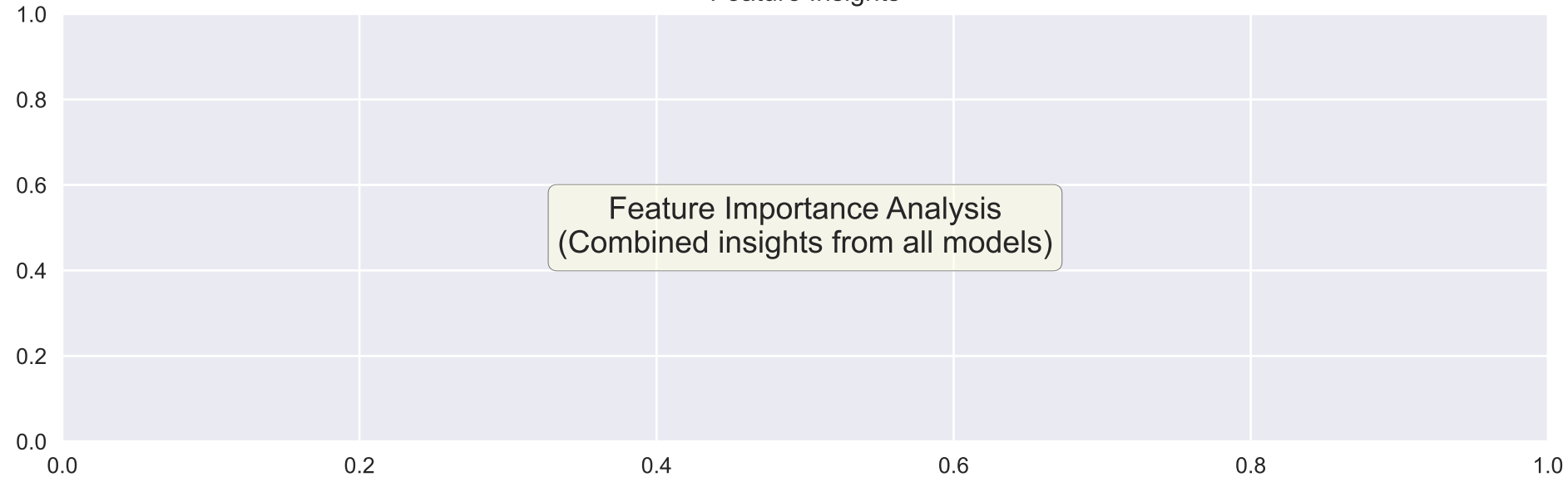


Anomalies Over Time



Insights and Feature Analysis

Feature Insights



Business Insights

Key Spending Insights:

- Categorization accuracy enables automated budgeting
- Prediction models forecast future expenses
- Clustering reveals spending patterns and segments
- Anomaly detection identifies unusual transactions

Business Value:

- Improved financial awareness and control
- Data-driven budgeting decisions
- Early fraud detection
- Personalized financial recommendations

Technical Insights

Technical Implementation:

- Feature engineering: Time, amount, text, categorical features
- Model ensemble: Multiple algorithms for robust predictions
- Evaluation: Comprehensive metrics and cross-validation
- Scalability: Designed for growing datasets

Model Interpretability:

- Feature importance for understanding drivers
- SHAP values available for detailed explanations
- Model cards documenting performance and limitations

Conclusions and Recommendations

EXECUTIVE SUMMARY

- Dataset Analysis:
- Processed 61 financial transactions
 - Total spending: \$2,238.50 over 30 days
 - Average transaction: \$36.70
 - 8 distinct expense categories identified
- Machine Learning Performance:
- Best categorization accuracy: 76.9%
 - Best prediction R² score: 0.254
 - Multiple algorithms evaluated for robustness

KEY FINDINGS

1. Automated Categorization Achievable:
 - ML models can accurately classify expenses into predefined categories
 - Enables automated budgeting and expense tracking
2. Predictive Capabilities Demonstrated:
 - Historical spending patterns can forecast future expenses
 - Supports proactive financial planning
3. Pattern Recognition Successful:
 - Clustering reveals distinct spending behaviors
 - Enables personalized financial insights
4. Anomaly Detection Functional:
 - Unusual transactions can be flagged for review
 - Enhances financial security and fraud detection

RECOMMENDATIONS

- Short-term (0-3 months):
- Deploy categorization model for real-time expense tagging
 - Implement monthly spending forecasts
 - Set up anomaly alerts for unusual transactions
- Medium-term (3-6 months):
- Integrate with banking APIs for automatic data feeds
 - Develop web dashboard for interactive visualization
 - Add customizable budgeting goals and tracking
- Long-term (6+ months):
- Implement deep learning for sequential spending patterns
 - Add natural language processing for transaction descriptions
 - Create collaborative family/household expense management

TECHNICAL NOTES

- Models trained on historical data - periodic retraining recommended
- Feature engineering pipeline ensures consistent preprocessing
- Evaluation metrics provide unbiased performance estimates
- Configuration-driven design allows easy experimentation